

ENTITY RELATIONSHIP DIAGRAMS

YESE

OpenTek School Enterprise Management System (SEMS-ERP)

Modular Data Model — Domain Map, per-Domain ERDs & Data Dictionary

Version 1.0 — Aligned to the Master Implementation Prompt & SRS

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Table of Contents

1. Introduction

1.1 Purpose

This document presents the data model for YESE — OpenTek's multi-tenant, modular School Enterprise Management System (SEMS-ERP). Because the platform is genuinely modular, the model is presented as a modular set: a domain map showing how the domains relate, followed by a detailed entity-relationship diagram for each domain, and a consolidated data dictionary. It is derived from the database architecture in the Master Implementation Prompt and the SRS, and targets PostgreSQL.

1.2 Notation & Conventions

- **Notation:** crow's-foot ERD. Each entity lists its columns with markers — **PK** (primary key), **FK** (foreign key), **UQ** (unique). A leading * marks a required (NOT NULL) column.
- **Database:** PostgreSQL. This resolves the earlier Postgres-vs-MongoDB ambiguity decisively in favour of PostgreSQL (relational integrity, row-level tenant isolation, financial ledgers).
- **Multi-tenancy:** every tenant-owned table carries a `tenant_id` foreign key; isolation is enforced at the data-access layer, never the frontend.
- **Data ownership:** each domain owns its tables. Cross-domain references use foreign keys to another domain's primary key, but a module never manipulates another module's internal tables directly.
- **Identifiers:** UUID surrogate keys throughout, for global uniqueness across tenants.

2. Domain Map (Cross-Domain Relationships)

The domain map shows the platform's domains and how they depend on one another. Every domain depends on the Core Platform for tenancy, identity, RBAC and module activation. Arrows indicate dependency and data reference between domains, not table-level foreign keys (those appear in the per-domain diagrams that follow).

YESE — Domain Map & Cross-Domain Relationships

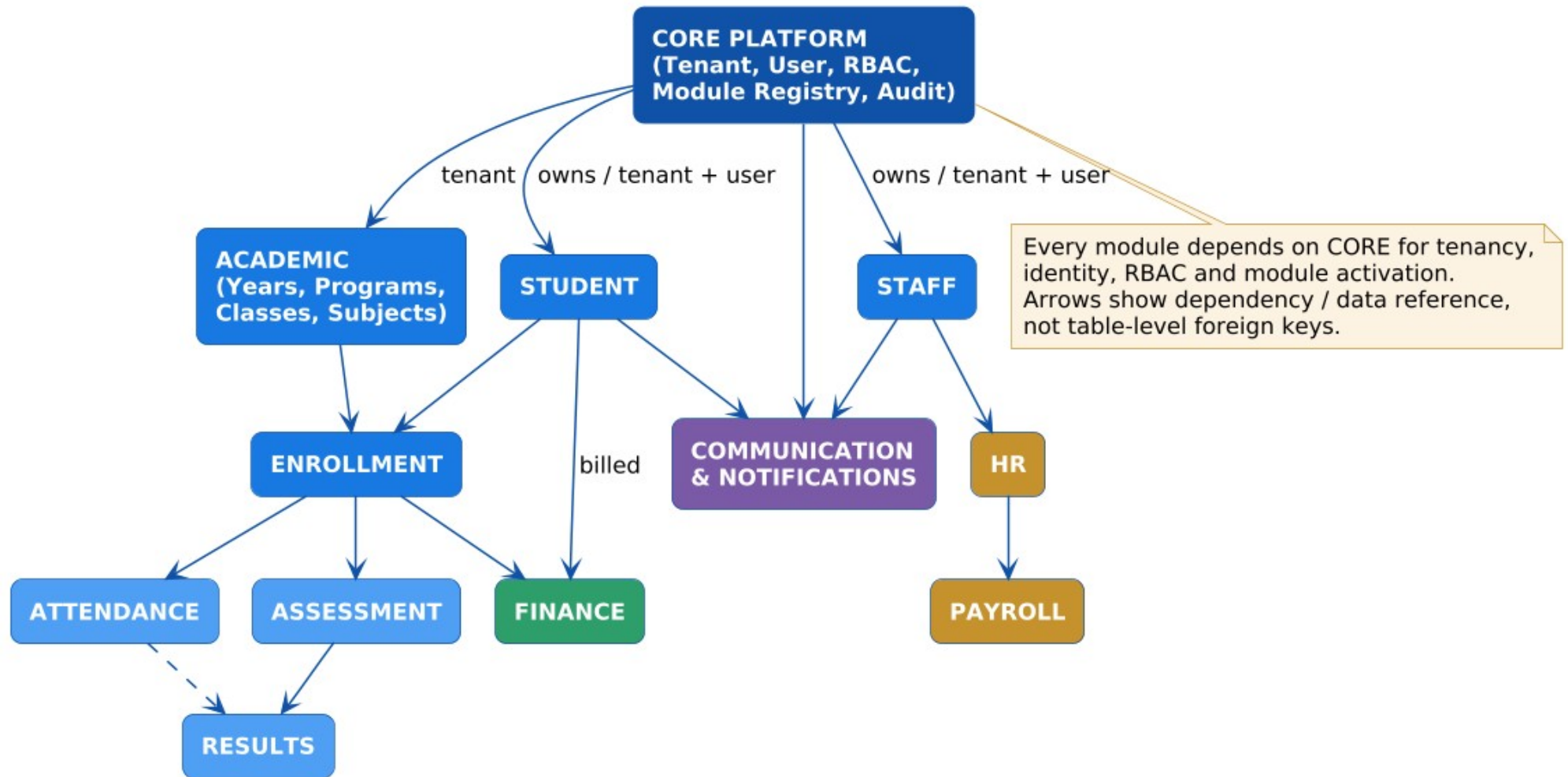


Figure 1 — YESE domain map.

3. Core Platform Domain

The Core Platform underpins every other module: multi-tenancy (Tenant), identity and accounts (User, Identity), the two-layer RBAC model (Role with fixed archetypes plus tenant-defined custom roles, Permission, and the User-Role / Role-Permission junctions), the Module Registry that makes the platform configurable (Module, Module_Version, Module_Dependency, Tenant_Module, Module_Configuration), and Audit_Log for traceability.

YESE — Core Platform Domain (Tenancy · Identity · RBAC · Modules · Audit)

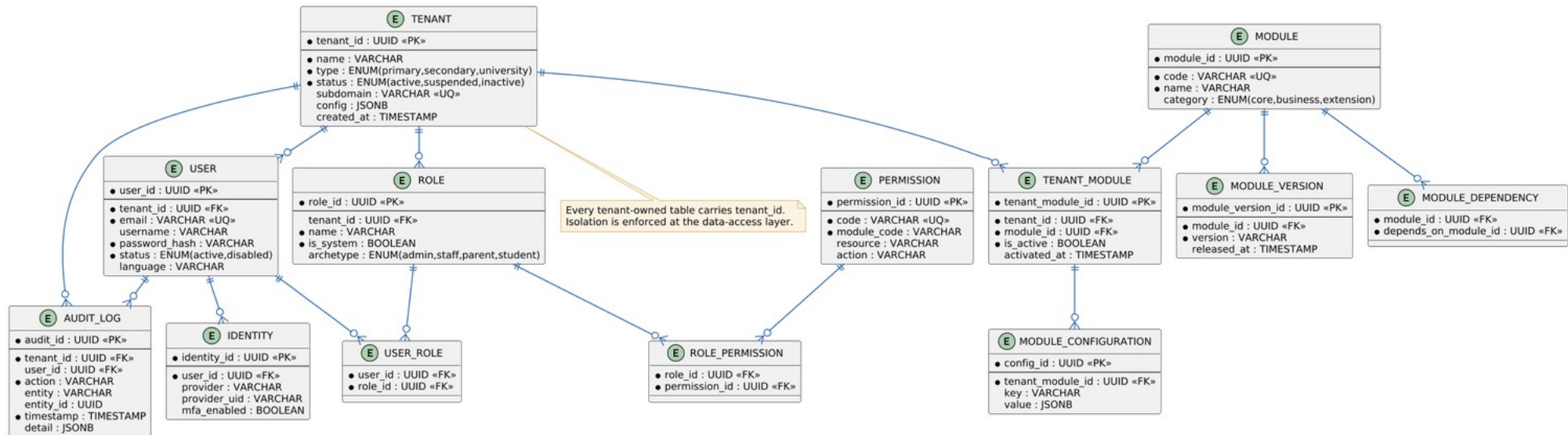


Figure 2 — Core Platform domain ERD.

4. Student & Staff Domains

The Student domain holds student records, profiles, statuses, documents, and the many-to-many Student-Guardian relationship (a student may have several guardians and a guardian several children). Not every student has a user account — primary pupils may have none. The Staff domain mirrors this for teaching and non-teaching staff, with types, statuses, profiles and assignments.

YESE — Student & Staff Domains

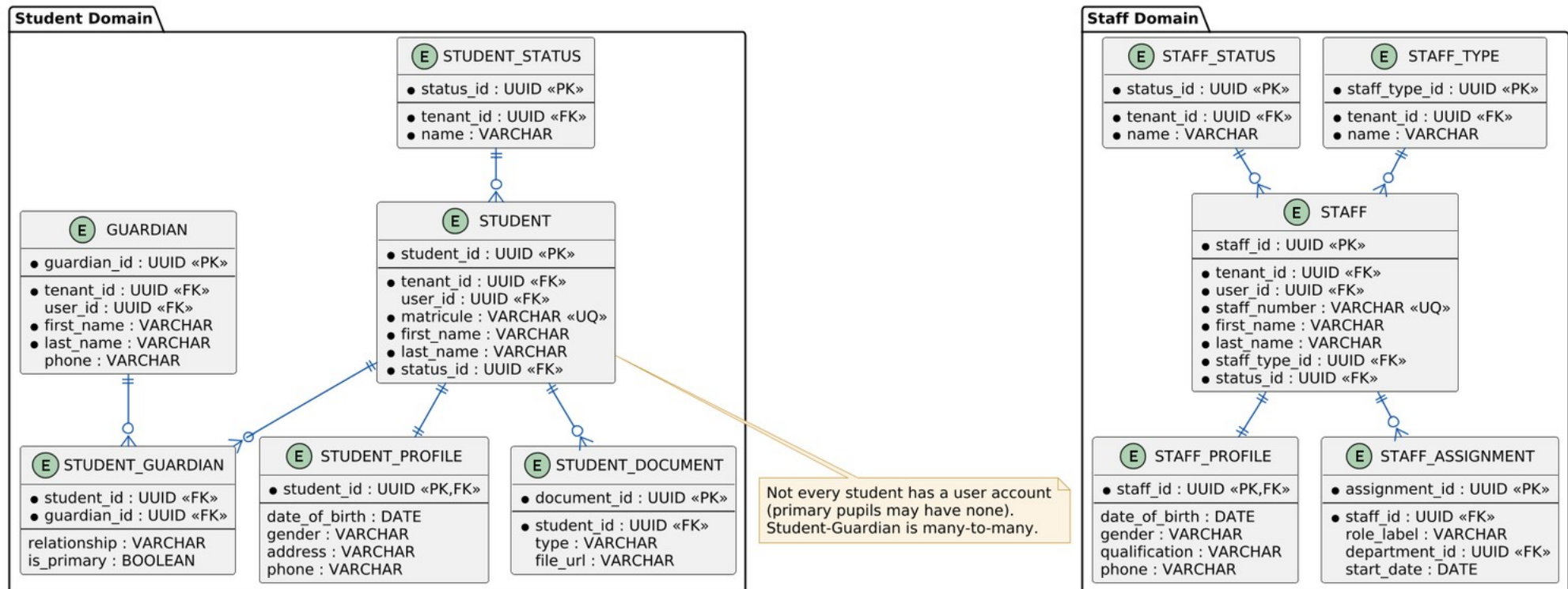


Figure 3 — Student & Staff domains ERD.

5. Academic Structure & Enrollment

The Academic domain defines the institution's academic frame: academic years and terms, departments, programs, classes and sections, subjects and courses (with credit units for universities), curricula, and teacher assignments. The Enrollment domain places a student into a section for an academic year; Teacher_Assignment binds staff to a subject within a section.

YESE — Academic Structure & Enrollment Domains

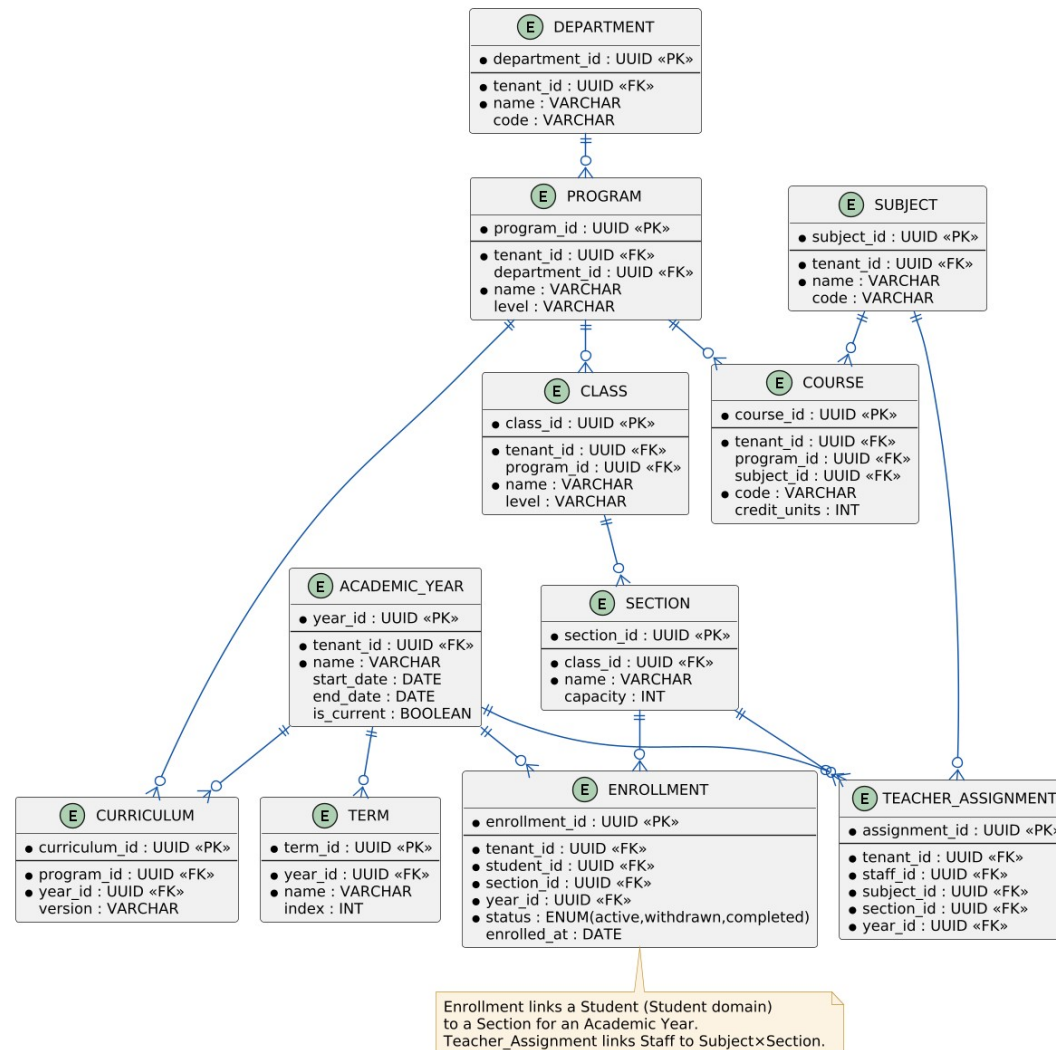


Figure 4 — Academic structure & enrollment ERD.

6. Academic Operations (Attendance · Assessment · Results)

Academic operations build on enrollment. Attendance records per-session student presence against configurable statuses. Assessment captures assessments, their components, per-student attempts, scores and grade entries, with a grade lifecycle of entered → validated → locked. Results consolidate grades into results, summaries, report cards (school) and transcripts (university).

YESE — Academic Operations (Attendance · Assessment · Results)

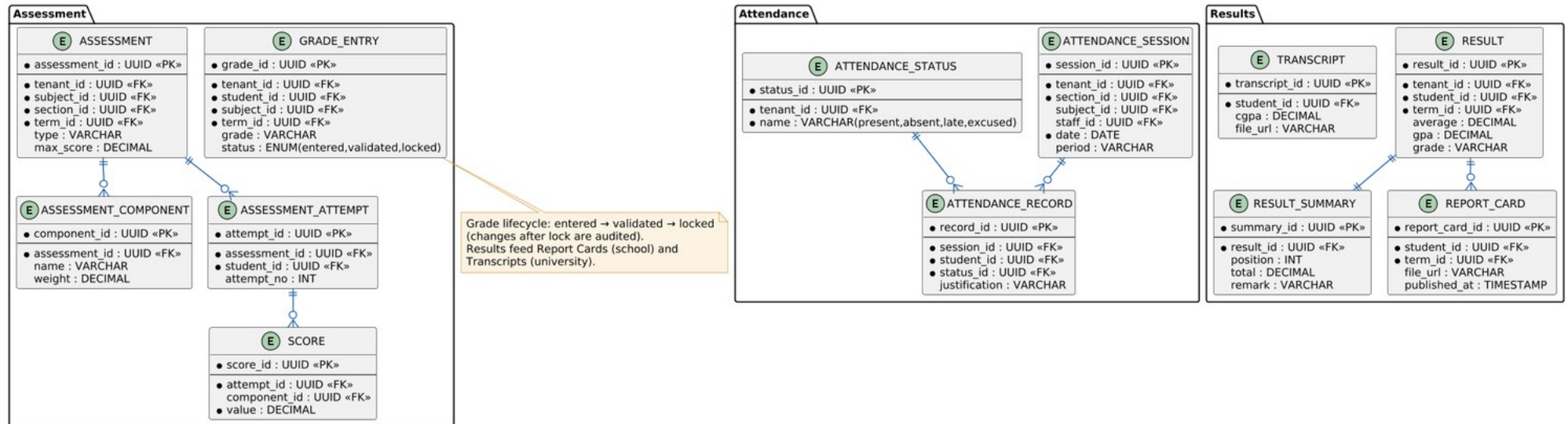


Figure 5 — Academic operations ERD.

7. Finance Domain

The Finance domain defines fee structures and items, generates invoices and invoice items, records payments (including Mobile Money channels) and issues uniquely-numbered receipts, and maintains an immutable financial-transaction ledger with discounts and scholarships. Financial records are never deleted — corrections use void, reversal or cancellation with an audit trail.

YESE — Finance Domain

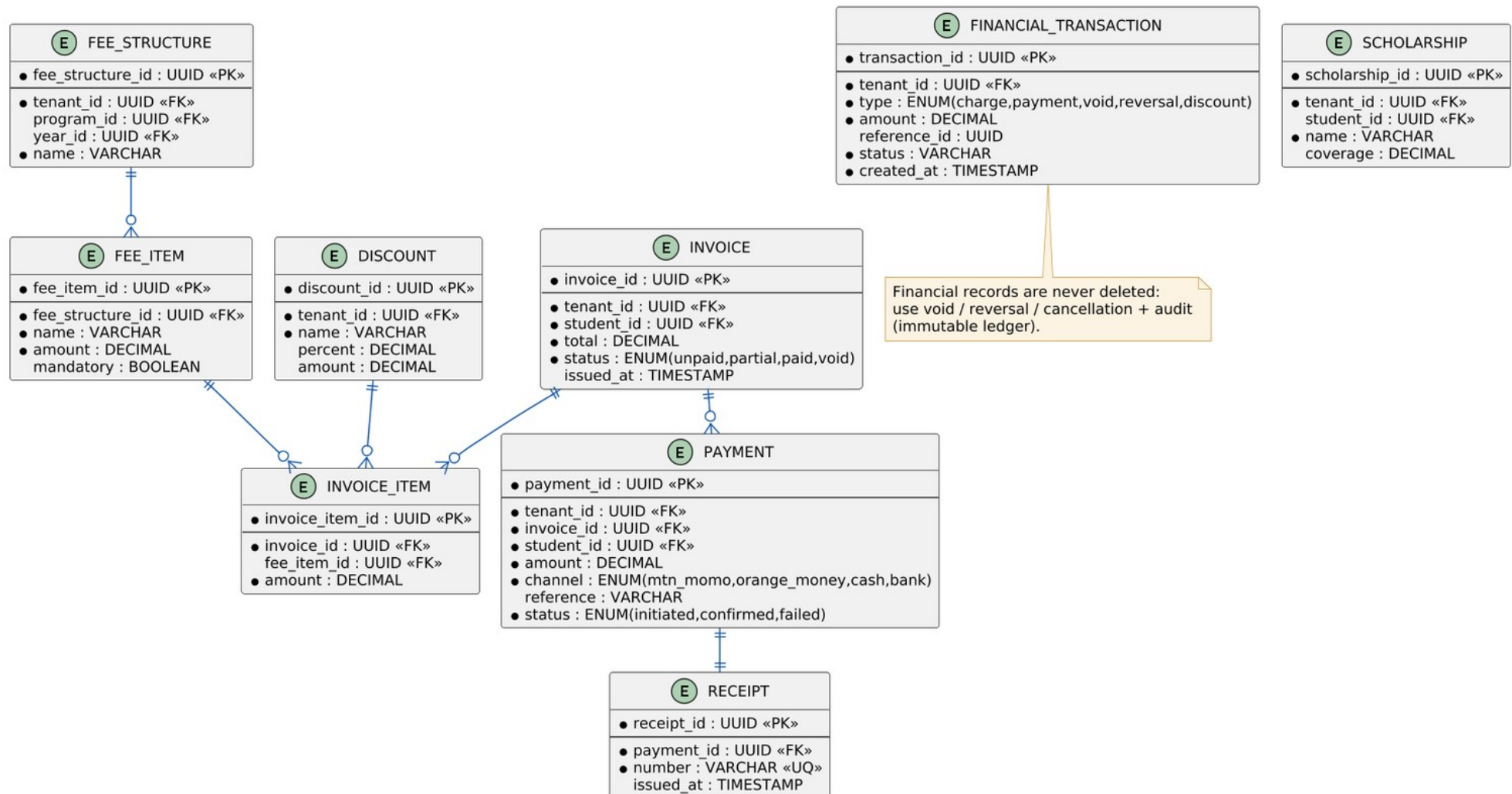


Figure 6 — Finance domain ERD.

8. HR & Payroll Domains

The HR domain manages leave types and requests with approvals, staff attendance and staff performance. The Payroll domain manages salary structures, payroll runs and per-staff payroll items with their deductions, bonuses and generated payslips. Both depend on the Staff domain.

YESE — HR & Payroll Domains

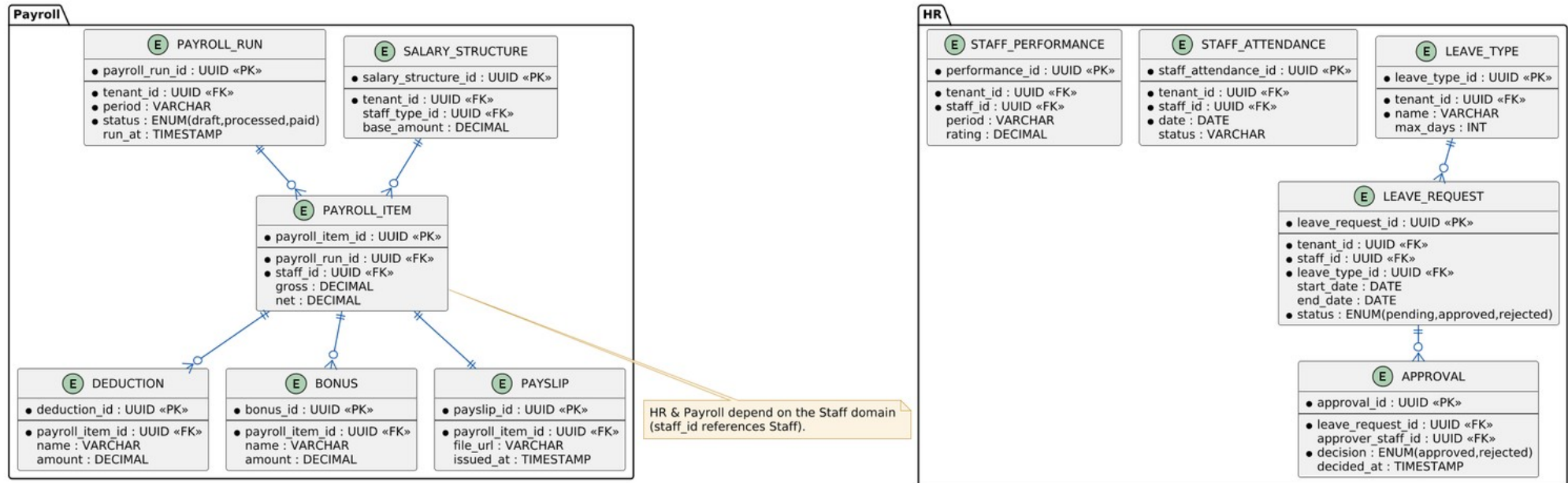


Figure 7 — HR & Payroll domains ERD.

9. Communication & Notifications

The Communication domain models conversations and their participants, messages, announcements, and — critically — `Communication_Rule`, which encodes the communication firewall enforced at the backend (for example, Parent → Student and Student → Admin direct messaging are blocked). The Notifications domain records notifications and their multi-channel delivery (in-app, email, SMS, WhatsApp).

YESE — Communication & Notifications Domains

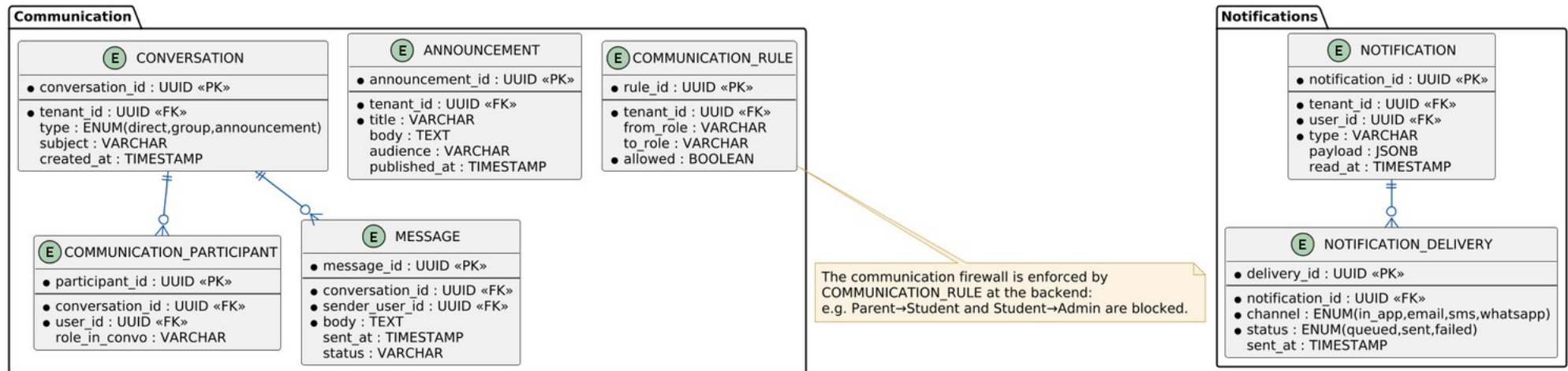


Figure 8 — Communication & Notifications ERD.

10. Data Dictionary

A consolidated reference of the principal entities by domain. Tenant-owned entities additionally carry `tenant_id` (omitted below for brevity).

10.1 Core Platform

Entity	Purpose
Tenant	One institution (tenant); root of all tenant-scoped data.
User	An account belonging to a tenant.
Identity	Authentication identity/provider linkage and MFA for a user.
Role	A role — a fixed archetype (admin/staff/parent/student) or a tenant-defined custom role.
Permission	A discrete permission (module · resource · action).
User_Role / Role_Permission	Junctions assigning roles to users and permissions to roles.
Module	A registered platform module.
Module_Version / Module_Dependency	Module version metadata and inter-module dependencies.
Tenant_Module	Per-tenant activation of a module (the modularity engine).
Module_Configuration	Per-tenant module settings.
Audit_Log	Immutable record of critical actions.

10.2 Student & Staff

Entity	Purpose
Student / Student_Profile / Student_Status	Student record, extended profile, and lifecycle status.
Guardian / Student_Guardian	Guardian record and the many-to-many student-guardian link.
Student_Document	Documents attached to a student.
Staff / Staff_Profile / Staff_Type / Staff_Status	Staff record, profile, type and status.
Staff_Assignment	A staff member's assignment (e.g. to a department).

10.3 Academic & Enrollment

Entity	Purpose
Academic_Year / Term	The academic calendar.
Department / Program / Class / Section	The structural hierarchy of the institution.
Subject / Course / Curriculum	Subjects, credit-bearing courses and program curricula.
Teacher_Assignment	Binds a staff member to a subject within a section for a year.
Enrollment	Places a student into a section for an academic year.

10.4 Academic Operations

Entity	Purpose
Attendance_Session / Attendance_Record / Attendance_Status	Attendance per session, per student, with status.
Assessment / Assessment_Component	An assessment and its weighted components.
Assessment_Attempt / Score	A student's attempt and the scores recorded.
Grade_Entry	A validated/locked grade for a student in a subject and term.
Result / Result_Summary / Report_Card / Transcript	Consolidated results, summaries and generated documents.

10.5 Finance

Entity	Purpose
Fee_Structure / Fee_Item	Fee definitions and their line items.
Invoice / Invoice_Item	A student's invoice and its lines.
Payment / Receipt	Payments (incl. Mobile Money) and issued receipts.

Entity	Purpose
Financial_Transaction	Immutable ledger entry (charge/payment/void/reversal/discount).
Discount / Scholarship	Fee reductions and scholarship coverage.

10.6 HR & Payroll

Entity	Purpose
Leave_Type / Leave_Request / Approval	Leave configuration, requests and approvals.
Staff_Attendance / Staff_Performance	Staff attendance and performance records.
Salary_Structure	Salary basis per staff type.
Payroll_Run / Payroll_Item	A payroll cycle and its per-staff items.
Deduction / Bonus / Payslip	Payroll item deductions, bonuses and generated payslips.

10.7 Communication & Notifications

Entity	Purpose
Conversation / Communication_Participant / Message	Threads, their participants and messages.
Announcement	Broadcast announcements to an audience.
Communication_Rule	Backend-enforced messaging firewall rules.
Notification / Notification_Delivery	Notifications and their multi-channel delivery.

— End of YESE Data Model (ERDs & Data Dictionary), Version 1.0 —